

# William Liu

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High school senior interested in solving challenging problems, building creative projects, and exploring advanced maths and sciences. Aspiring to become a scientist or engineer to help create a better shared future.

## Education

**Ronald Reagan High School** - 12th grade, expected graduation May 2027

- **Class Rank:** 3/849 (Unweighted GPA 99.2/100, Weighted GPA 108.2/100)
- **Test Scores:** SAT: 1600 | AIME: 6 | AMC 12: 108
- **Relevant Coursework:** Multivariable Calculus, AP Physics C: Electricity/Magnetism, AP CompSci A, AP Chemistry

## Experience

**Welch Summer Scholar, Henkelman Lab, UT Austin**

*June 2026 - Present*

- Accepted into selective summer chemistry research program (7% acceptance rate, one of 43 Scholars)
- Engineered a Python-based materials screening pipeline using machine learning interatomic potentials (MLIPs) to accelerate discovery of electrode materials
- Screened 150,000+ candidate materials using MLIPs, and used results to train a generative AI model to propose novel electrode materials for lithium extraction
- Co-authoring a publication documenting novel materials discovery methodology and results

**Independent Programmer**

*August 2023 - Present*

- Built a school bell schedule app with Svelte and TailwindCSS used by 30+ students
- Created an app to calculate and visualize a Fourier Transform with epicycles
- Created and maintained a webpage for school's UIL Academics team to recruit members and share resources
- Designed and deployed a customized personal website to showcase portfolio of projects and technical skills

**Reagan Mu Alpha Theta**

*VP of Academics (April 2026 - Present), Inducted Member (October 2023 - Present)*

- Led math competitions, test prep sessions, Estimathons, and other math-related events for 80+ members
- Founded club's math team and designed advanced curriculum to prepare students for competition mathematics

**Reagan Orchestra**

*Violinist (August 2023 - Present)*

- Auditioned into varsity Chamber Orchestra and All-Region Orchestra multiple times
- Led 2nd violin section of varsity orchestra; made section-wide musical decisions and coordinated with director and other section leaders to prepare for concerts and competitions
- Performed for 5,000+ audience members in school musicals, seasonal concerts, and community outreach performances

## Skills

- **Programming Languages:** Python, Java, JavaScript, HTML/CSS, C++
- **Programming Frameworks:** Pandas, NumPy, PyTorch, Svelte, matplotlib, TailwindCSS, SQLite
- **Other Interests:** Playing violin and piano, running, swimming, reading, thinking about public transit, collecting maps

## Awards

- American Invitational Mathematics Examination (AIME) qualifier (2024-2026)
- USA Computing Olympiad (USACO) Silver Division qualifier
- American Mathematics Competitions (AMC) 1st Place, Reagan High School (2023-2025)
- Stanford Math Tournament 4th place Team Round with the San Antonio Math Circle (2026)
- UIL Math State qualifier, Science/Computer Science Regional qualifier (2024-2026)
- Student Volunteer Service Award - Gold Level (2025, 2026)
- National Merit Semifinalist (2026)